

Thermodynamic State Vector Baseline Structurer: Architectural Foundations and Implementation in Sprint 027

Lead Scientific Communications & Academic Outreach Agent

Web of Life Project

<https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 027 Report

Abstract

This preprint details the architectural design and implementation of Sprint 027 within the Web of Life project, focusing on the Thermodynamic State Vector baseline structurer (`src/thermodynamics/state_vector.ts`). By codifying baseline ambient temperatures ($T_0 = 288.15$ K) and structuring discrete surface energy flux records (shortwave solar radiation, longwave thermal emission, latent heat, and sensible heat), the module ensures rigorous adherence to the First and Second Laws of Thermodynamics. We present the governing mathematical formulations for energy conservation, entropy generation, and monad-based stock transfer equations that maintain physical consistency across simulated Earth Pod subsystems.

1 Introduction and Thermodynamic Foundations

The Web of Life simulation framework models complex ecological interactions and planetary biogeochemical cycles. To prevent unphysical energy generation or entropy destruction, all ecosystem state changes must be governed by rigorous thermodynamic laws. Sprint 027 introduces the **Thermodynamic State Vector**, acting as an immutable state container for temperature, energy fluxes, cumulative entropy, and simulation epochs.

1.1 First Law of Thermodynamics (Energy Conservation)

The net energy balance across a system boundary layer is defined by radiative and convective surface fluxes:

$$\Delta E_{net} = R_{sw}^{\downarrow} - (R_{lw}^{\uparrow} + H_{latent} + H_{sensible}) \quad (1)$$

where:

- $R_{sw}^{\downarrow}$: Incoming shortwave solar radiation flux (W/m^2)
- $R_{lw}^{\uparrow}$: Outgoing thermal longwave emission flux (W/m^2)
- H_{latent} : Latent heat flux due to evapotranspiration and phase change (W/m^2)
- $H_{sensible}$: Sensible heat flux due to convective heat transfer (W/m^2)

1.2 Second Law of Thermodynamics (Entropy Generation)

In accordance with the Second Law, incremental and cumulative entropy generation within bounded system transitions must remain non-negative:

$$dS_{gen} \geq 0 \implies S_{t+\Delta t} \geq S_t \quad (2)$$

2 Module Architecture and Implementation

The state vector is implemented in TypeScript within `src/thermodynamics/state_vector.ts`. It provides immutable data structures and builder functions to instantiate valid baseline states.

```
export interface FluxRecord {
  solarRadiation: number;    // Incoming shortwave flux (W/m^2)
  thermalEmission: number;   // Outgoing longwave flux (W/m^2)
  latentHeat: number;        // Evapotranspiration / phase change flux (W/m^2)
  sensibleHeat: number;      // Convective heat transfer flux (W/m^2)
}

export interface IThermodynamicStateVector {
  temperature: number;
  fluxes: FluxRecord;
  entropy: number;
  timestamp: number;
  clone(overrides?: ThermodynamicStateVectorOptions): IThermodynamicStateVector;
  validateFirstLaw(): boolean;
  validateSecondLaw(): boolean;
}
```

3 Monad Stock Transfer & State Evolution Equations

State transitions are evaluated via pure monad operations (`ThermodynamicMonadProcess.step`), updating thermal energy capacities, surface temperatures, and entropy generation metrics over time step Δt :

$$E_{t+\Delta t} = E_t + \Delta E_{net} \cdot \Delta t \quad (3)$$

$$S_{t+\Delta t} = S_t + \frac{|\Delta E_{net}|}{T} \cdot \Delta t \quad (4)$$

$$T_{t+\Delta t} = \max \left(0.1, T_t + \frac{\Delta E_{net} \cdot \Delta t}{C_{eff}} \right) \quad (5)$$

where $C_{eff} = 2.0 \times 10^5 \text{ J}/(\text{K} \cdot \text{m}^2)$ represents the effective surface thermal inertia parameter.

4 Conclusion and Repository Reference

Sprint 027 establishes the foundational thermodynamic accounting layer required for all subsequent biogeochemical and ecological simulations in Web of Life.

For complete source code, test suites, and ongoing development updates, please refer to the official repository:

<https://github.com/pascalranoroarijaona/WebOfLife>